

# Final Project

$$F = G \frac{m_1 m_2}{d^2}$$

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$F - E + V = 2$$

$$E = mc^2$$

$$ds \geq 0$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

Computação Gráfica / Computer Graphics  
(L.EIC, 3rd course year)

# General Objectives

- Apply **geometric modeling techniques** to create complex composite objects from simple primitives;
- Use **geometric transformations** (translation, rotation, scaling) and the composition of transformations to instantiate, shape, and animate scene elements;
- Create and configure **illumination, materials, and textures** to achieve visually realistic surfaces;
- Develop **custom GLSL shaders** for procedural animation and visual effects;
- Implement **real-time user interaction** through keyboard controls and GUI elements;
- **Animate scene** components using time-dependent functions;
- Practice **collaborative software development** with version control (Git tags, commits, branching).

# Theme

- The theme of this year's scene is a **spring prairie landscape**.
- And your challenge is to build **a game!**



Image generated by AI not depicting the final visual style...

# Level world

The scene is composed by:

- ground surface over an elevated terrain;
- On top, a cloudy sky;
- The ground contains multiple details and scattered elements.
- The two other relevant entities of the scene
  - A covered light wagon (wagon, aka prairie schooner)
  - A barn are the.

# Gameplay

The game is a survival challenge: keep the wagon driver (you) “alive” for as much time as possible!

- In the end, the score is the total number of seconds of a single gameplay.
- The barn is where the wagon starts, and it has a starting number of health points (HP).
- Health decreases per unit of time and increases with food.
- Hay bales provide food by granting HPs when delivered to the barn. (Hay bales appear randomly over the prairie)

Drive the wagon carefully along the path/fields, especially avoiding obstacles (e.g., rocks), as they can damage the wagon.

# Scene elements (1/4)

## **Sky, clouds, and sun**

- Huge half-sphere geometry, faces turned inward, inverted normals, panoramic landscape environment.
- Sun, Directional light/Spotlight

Advanced:

- Procedural clouds , animated cloud layer (as a second texture) near sky-sphere.

# Scene elements (2/4)

## **Terrain elevation**

- Keep subtle rolling hills (use height maps for this), which are typical of the prairie.

Advanced:

- Terrain using procedural generation.

## **Ground surface**

- Soil texture with dirt patches.
- Pathway for wagons.

Advanced:

- Smooth pathways (ex. cubic curve segment paths).

# Scene elements (3/4)

## **Scatter Elements**

- Rocks, multiple textures, and vertex perturbation.

Advanced:

- Procedural generation of items.

## **Flora**

- Parameter-based flowers.  
Parametrisation should foster natural randomness (textures, colours, scale, and flower placement around the terrain)

Advanced:

- procedural generation of several components of flora items (ex. flowers).

# Scene elements (4/4)

## Grass

- Dense grass patches
- Dead grass patches

## Advanced:

- Wind-reactive instances  
(Shader over 3D blade geometry  
+ vertex shader wind animation)

# Covered light wagon/prairie schooner

Create a hierarchical model of a wagon with the following mandatory components (from top to bottom):

- a cloth cover occupying half the wagon length (to see the hay bales):
- a wooden wagon bed;
- a front tongue to attach horses/mules;
- and 4 wheels.

Advanced:

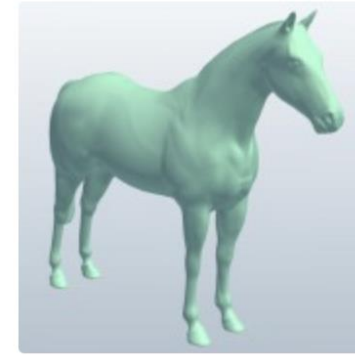
- Highly detailed (geometry and textures) wagon.

# The Prairie schooner



# Horses

- Horses/mules should be imported in OBJ format. (An extra WebCFG class is provided towards this purpose)
- No animation required.

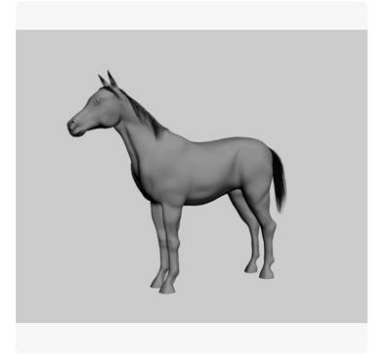


**American Paint Horse Neutral V1**

.obj .stl

**FREE**

35,682

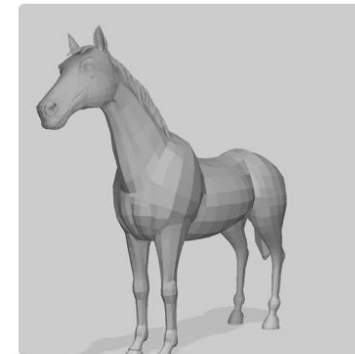


**Horse**

.max .3ds .fbx .obj

**\$3**

13,683

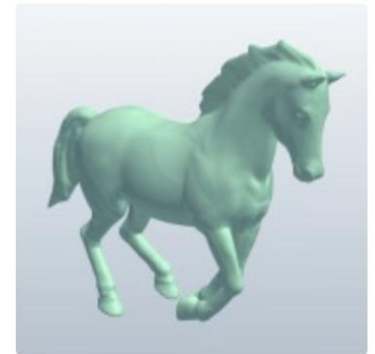


**Horse**

.ma .obj

**\$25**

5,500



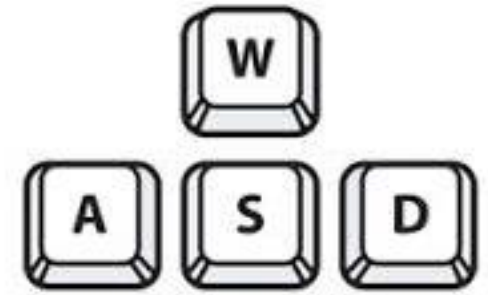
**Morgan Horse Galloping V1**

.obj .stl

**FREE**

8,655

# Wagon interaction mechanics



- Smooth keyboard control (W, A, S, D + P, L):
  - W accelerates forward (up to horse walk speed)
  - A steers front wheel left
  - D steers front wheels right
  - S reduces speed
  - P for hale pick-up
  - L for hale drop

- Forward movement only;

Advanced:

- acceleration using physics (kinematics).

# Barn

- Create a simple barn using a Cube with a wood plank texture.
- Roof composed of half a prism with a dark wood texture.
- Apply textures to mimic windows, doors, and other details.
- Bale circular delimited area in front of the barn. The circular area changes look depending on whether the wagon intersects it.

## Advanced:

- Highly detailed barn and circular area, and visual feedback.



# Shaders

Apply shaders to the following scene elements:

- **Grass moving with the wind;**
- Vertical pinpointing arrows.



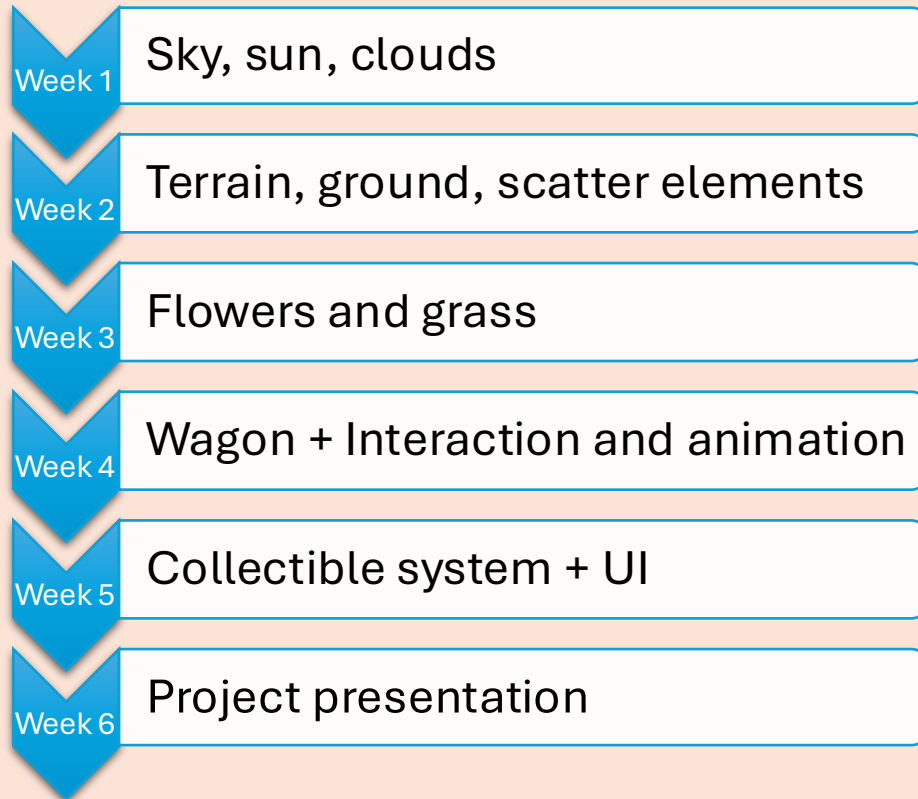
# Shaders

Apply shaders to the following scene elements:

- Grass moving with the wind;
- **Vertical pinpointing arrows.**



# Scheduled plan



|    |   |        |        |        |        |        |        |        |
|----|---|--------|--------|--------|--------|--------|--------|--------|
| 9  | I | 20-Apr | 21-Apr | 22-Apr | 23-Apr | 24-Apr | 25-Apr | 26-Apr |
| 10 | P | 27-Apr | 28-Apr | 29-Apr | 30-Apr | 1-May  | 2-May  | 3-May  |
|    |   | 4-May  | 5-May  | 6-May  | 7-May  | 8-May  | 9-May  | 10-May |
| 11 | I | 11-May | 12-May | 13-May | 14-May | 15-May | 16-May | 17-May |
| 12 | P | 18-May | 19-May | 20-May | 21-May | 22-May | 23-May | 24-May |
| 13 | I | 25-May | 26-May | 27-May | 28-May | 29-May | 30-May | 31-May |
| 14 | P | 1-Jun  | 2-Jun  | 3-Jun  | 4-Jun  | 5-Jun  | 6-Jun  | 7-Jun  |



# Evaluation criteria

Basic

0 - 40%

Good

40% - 60%

Advanced

60% - 80%

Expert

80% - 100%

| Criterion                                       | Weight |
|-------------------------------------------------|--------|
| 1. Explicit and procedural modeling of entities | 4.0    |
| 2. Lighting conditions, materials, and textures | 4.0    |
| 3. Overall scene and entities complexity        | 3.0    |
| 4. Animated entities                            | 3.0    |
| 5. Advanced effects through shaders             | 3.0    |
| 6. Aesthetics and creativity <sup>1</sup>       | 2.0    |
| 7. Software Quality <sup>2</sup>                | 1.0    |
| TOTAL                                           | 20.0   |

A black and white photograph of a person walking up a long, dark staircase. The person is silhouetted against a bright light source at the top of the stairs, creating a strong backlight effect. The staircase is flanked by dark walls and railings. In the background, a tall building with many windows is visible, also silhouetted against the bright light. The overall mood is one of challenge and perseverance.

# Challenges

Advanced topics and specific techniques